

Core Concepts in NLP

10 Foundational Questions and 1-Line Answers

1. Tokenization Strategies

Subword tokenization breaks rare words into smaller, recurring chunks to handle unknown vocabulary while keeping common words intact.

2. Stemming vs. Lemmatization

Stemming crudely chops off word endings for speed, whereas lemmatization uses a dictionary to accurately return a word's proper linguistic base form.

3. TF-IDF (Vectorization)

The Inverse Document Frequency (IDF) mathematically penalizes overly common words (like "the") so that rare, highly informative keywords stand out.

4. Word Embeddings

Contextual embeddings dynamically change a word's mathematical vector based on its surrounding sentence, effectively resolving words with multiple meanings.

5. The Attention Mechanism

Self-attention processes all words simultaneously to calculate their relevance to each other, solving the long-term memory loss issues of sequential RNNs.

6. Named Entity Recognition (NER)

NER identifies and categorizes specific proper nouns (like people, places, and organizations) to transform raw text into structured, searchable data.

7. Syntactic Ambiguity

Because words change parts of speech depending on usage (e.g., "book" a flight vs. read a "book"), models must analyze surrounding context to assign correct grammatical tags.

8. N-gram Language Modeling

As 'N' increases, the model gains better context but suffers from severe data sparsity, meaning exact matches for long word sequences become extremely rare.

9. Transformer Architecture Basics

Because Transformers process all words in parallel without inherent sequence, positional encodings are mathematically injected to tell the model the exact order of the words.

10. Model Evaluation

Perplexity measures how confused a model is by new text, meaning a lower score is always better because it indicates high confidence and prediction accuracy.